

Temperature Dependent Space Charge and Breakdown Strength of PP/ULDPE/Graphene Nanocomposites for HVDC Extruded Cable Insulation

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ABSTRACT

The space charge behavior, conductivity and breakdown strength are critical issues in the design of insulation for high voltage direct current (HVDC) extruded cables, which is greatly affected by the operating temperature at different load conditions. The space charge, DC conductivity and breakdown characteristics of polypropylene (PP) based insulation at various temperatures are investigated in this paper. The space charge, DC conductivity and the breakdown strength of pure PP, PP/ultralow density polyethylene (ULDPE) blends and its graphene nanocomposites were examined at 30, 60 and 90 °C. The results show that, at elevated temperature, the homo-charge injection in PP from both electrodes is enhanced. Significant charge accumulates in PP/ULDPE blends in the vicinity of the anode at 90 °C. A small amount (0.01 wt%) of graphene nanoplatelets in PP/ULDPE blends inhibit the injection of homo-charge at elevated temperatures. No dramatic change in the breakdown strength with increasing temperature is observed. The DC conductivity increases with increasing temperature, and the conductivity significantly increases from 60 to 90 °C compared to from 30 to 60 °C. The PP/ULDPE blends with 0.01 wt% graphene nanoplatelets exhibit good properties for potential HVDC cable insulation.

Index Terms — HVDC cable insulation, space charge, breakdown strength, high temperature, polypropylene, ULDPE, graphene nanoplatelets

1 INTRODUCTION

POLYPROPYLENE (PP) with good electrical insulation characteristics, non-crosslinking and recyclable properties, has drawn much attention as a potential high voltage direct current (HVDC) cable insulation [1,2]. Especially, the good thermostable performance of PP will be beneficial for a high capacity electric power cable transmission [3]. Although the inherent brittleness of PP could be a limitation for its application, blending with thermoplastic elastomer (TPE), has revealed one of the most effective and convenient methods for PP toughening [4,5]. Among the TPEs, the ultralow density polyethylene (ULDPE), an effective impact modifier for rigid matrices, has shown a good thermal and mechanical performance in PP/ULDPE blends[6].

The operating temperature of HVDC cables is nominally designed up to 90 °C, which varies with the load conditions for cables in operation [7]. The temperature variation along with the space charge, DC conductivity and breakdown characteristic in insulation are believed to be key issues for the

successful design and application of HVDC cable insulation [8-10]. The high density of space charge in the insulation, either in transit or trapped, cause a distortion of electric field, accelerate ageing, and increase the rate of electrical tree formation, eventually leading to breakdown [11,12]. At 90 °C, the space charge effects can be worse and more complicated [13-15]. In a DC cable, the electrical resistivity determines the distribution of electric field, which varies with temperature gradient, which comes about due to different loading conditions. When the temperature from the conductor to the sheath rises to an appreciable degree, the field distribution could be inverted, which results in an enhanced field or even a breakdown[7, 16]. In addition, since the physical property of the polymer will change at different temperatures; the temperature dependence of breakdown strength is of prime importance for the polymer insulation performance [15-18]. Therefore, the effects of different temperatures on space charge behavior, DC conductivity and breakdown strength of PP based polymer insulation has a great practical significance and research value for cable insulation.

Much research work has revealed that the introduction of nanoparticles can suppress the space charge accumulation and

enhance the electrical properties of polymer insulation, which is attributed to the charge-limiting effects of interfacial regions between the nanofillers and polymer phases [14, 19-20]. Although, some research work about the dielectric properties of PP and its nanocomposites have been done, most of this work has been done at room temperatures [2, 5, 18]. Recent research reported that a small addition of graphene nanoplatelets in polymers form numerous interaction zones and create many deep traps, for its specific structure and enormous surface area [19, 21]. This will do good for investigating the potential of PP based nanodielectrics at different temperatures.

In this paper, the space charge behavior, DC conductivity and breakdown strength of pure PP, PP/ULDPE blends and its nanocomposites with 0.01 and 0.05 wt% amount of graphene nanoplatelets were studied at 30, 60 and 90 °C. The temperature effects on charge injection and transit, dielectric properties of PP/ULDPE/graphene nanocomposites are discussed on the basis of experimental results.

2 EXPERIMENTAL PROCEDURES

2.1 PREPARATION OF SPECIMENS

The PP (F401) was supplied by Liaoning Huajin Chemical Engineering (Group) Co., Ltd., China, of which the density is 0.92 g/cm³. The ULDPE (ATTANE™ 4203) was supplied by Dow Chemical Company, of which the density is 0.905 g/cm³. The graphene nanoparticle with a diameter of 0.2~10 μm and a thickness of ~1 nm is from Tanfeng Tech. Inc, China.

The preparation of different specimens follows the process bellow. Certain proportions of PP, ULDPE, graphene nanoparticles were blended by two roll mill at 190 °C, with the rotation speed of 45 rpm for 10 min. A moderate amount weight of these blends were put in the stainless steel mold at the temperature of 190 °C and the pressure of 25 MPa for 10 min. Afterwards, they were cooled to 160 °C under the same pressure. With different stainless steel molds, some composites were made into a square of 8 cm length, with thickness of ~180 μm for space charge and DC conductivity test, and some were made with the thickness of ~70 μm for DC breakdown test. Details of the specimens prepared in this paper are shown in Table 1, and the corresponding code names with different amounts of graphene are also listed below.

Table 1. Specimens with different amounts of graphene.

Code name	PP (wt%)	ULDPE (wt%)	Graphene (wt%)
PP	100	0	0
P15	85	15	0
G1	85	14.99	0.01
G5	85	14.95	0.05

2.2 EXPERIMENTAL SETUP

The space charge distribution measurements of PP/ULDPE/graphene nanocomposites were tested under 60 kV/mm at 30, 60 and 90 °C, by pulsed electro-acoustic (PEA) method. The PEA equipment (supplied by Shanghai Xiangtie

Electromechanical Equipment Co., Ltd., China) has been designed for high temperature testing, with a heater and temperature thermocouple attached to the lower electrode (aluminum). The specimens were held between the two electrodes of the PEA system at the preset temperature for 5 min before voltage was applied. The voltage-on test (under the electric field) and voltage-off test (short-circuit) are performed for 60 and 10 min, respectively.

The DC conductivity of different specimens were measured by a 3-electrode system [5]. The electrode system was set in an oven and warmed before the test. The DC conductivity characteristics at different temperatures were measured based on the charging current, which was measured by an electrometer (Keithley 6517B) for 1200 s under an electrical field of 20 kV/mm at 30, 60 and 90 °C. The DC conductivity (σ) is calculated from the following equation:

$$\sigma = \frac{I}{V} \frac{4t}{\pi(d+g)^2} \quad (1)$$

where V is the DC voltage performed by anode, I is the recorded current, t is the thickness of the sample, d is the diameter of the cathode, and g is the distance between the cathode and guard ring.

The DC breakdown test is performed by means of a DC ramp test with two stainless steel electrodes [5], at 30, 60 and 90 °C. The electrodes and specimens are put in a glass container, full of insulating oil to avoid surface flashover during the test. Before the test, the oil and the electrodes were preheated by a heater attached to the outer bottom of the container, and the temperature value were set at 30, 60 and 90 °C. Under the voltage increasing rate of 500 V/s, the specimen is tested until the specimen breaks down. The experiment was carried more than 10 times for each sample and the two-parameter Weibull distribution is employed to characterize the DC breakdown strength.

3 EXPERIMENTAL RESULTS

3.1 SPACE CHARGE BEHAVIOR

As shown in Figures 1 and 2, under the voltage-on or voltage-off conditions, the time-dependent space charge distributions of the polarity and amount in the PP/ULDPE/Graphene nanocomposites vary with the sample formulation and test temperature. Figure 1 shows the space charge distributions during the voltage-on time of 60 min under 60 kV/mm and Figure 2 shows the space charge depolarization (voltage-off) after applying 60 kV/mm for 3600 s, both performed at 30, 60 and 90 °C.

In terms of the pure PP, it can be observed that only a small quantity of hetero-charge appears near the electrode during the voltage-on time at 30 °C. As shown in Figure 1a-1, the total charge injected into PP is gradually increasing during the polarization time. As shown in Figure 1b-1, when the test temperature rises to 60 °C, the injection of homo-charge from both electrodes is enhanced and there is a small amount of negative charge accumulation within the bulk near the

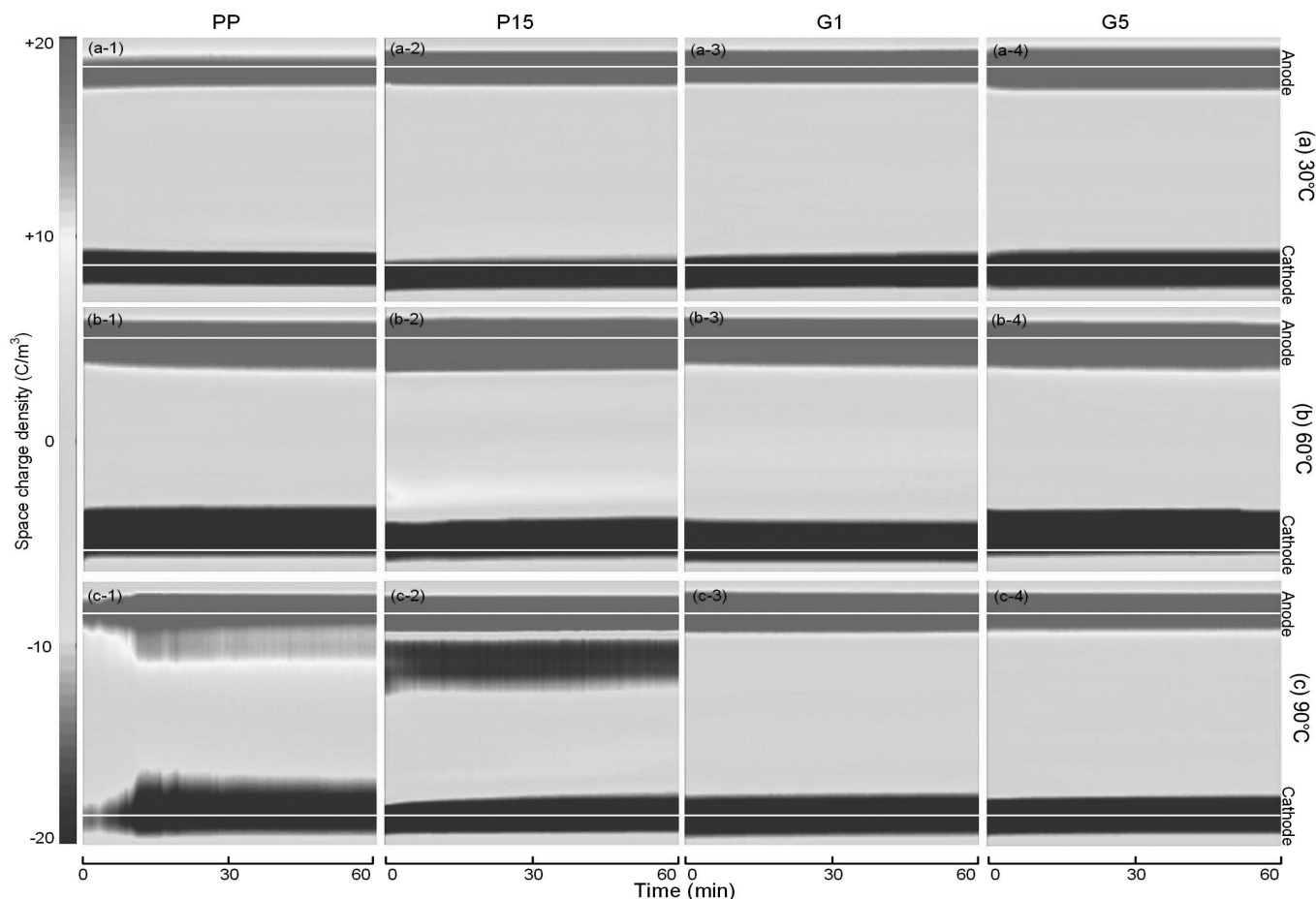


Figure 1. Space charge distribution in PP and PP/ULDPE/graphene nanocomposites under 60 kV/mm for 60 min at different temperature: (a) 30 °C, (b) 60 °C and (c) 90 °C.

cathode. When it reaches 90 °C (shown in Figure 1c-1), the homo-charge injection of PP has been improved largely from both electrodes, and there is a prominent space charge injection at the first 15 min. Then, the injected homo-charge decreases slowly and reaches a relative stable state, which still keeps a considerable quantity at the end of the test.

Compared to PP at the same test temperature, the space charge behavior of P15 shows a very different characteristic. As shown in Figure 1a-2, a small amount of negative charges injected from the cathode and the amount of the charges increases slowly with voltage-on time at 30 °C. As for the P15 at 60 °C, the hetero-charge appears adjacent to both electrodes. Compared with the positive charge near the cathode, it shows that a larger amount of negative charge accumulated near the anode in Figure 1b-2 during the voltage applied. However, when the test temperature reaches 90 °C, the negative charge becomes dominant near the anode and there is only a very small amount of positive charge accumulation in the vicinity of the cathode (shown in Figure 1c-2).

As for the PP and ULDPE blends with very low amounts of graphene nanoplatelets of 0.01wt%, there are much fewer charges injected from electrodes into the bulk of G1 in Figure 1a-3 at 30 °C. The space charge distribution of G1 at 60 °C shows that the homo-charge injection is strengthened and a small number of positive charge appears within the specimen

(shown in Figure 1b-3). As depicted in Figure 1c-3, when the test temperature reaches 90 °C, there are only low amounts of negative charge accumulation within the bulk near the anode with the stressing time.

With regard to the space charge performance of G5 at 30 °C from Figure 1a-4, it appears that there are also much fewer charges accumulated in sample when compared with G1 at the same temperature. The homo-charge injection is also enhanced in G5 at 60 °C, and a tiny amount of negative charge is dominant in the sample (shown in Figure 1b-4). However, as shown in Figure 1c-4, there is almost no charge accumulated in G5 at 90 °C.

As shown in Figures 2a-1, 2b-1 and 2c-1, the space charge distribution during short circuit condition of PP are at 30, 60 and 90 °C, respectively. In Figure 2a-1, the hetero-charge accumulated near the anode and cathode decrease slowly with the time and the negative charge induced by the positive charge in the bulk near the cathode decrease slowly. As shown in Figure 2b-1, the injected homo-charge can be found in the vicinity of both electrodes, and the negative charge near the cathode dissipates more slowly in comparison with the positive charges. As for Figure 2c-1, there are much more homo-charge injected from both electrodes at the beginning of 60 s, but they both dissipate very quickly.

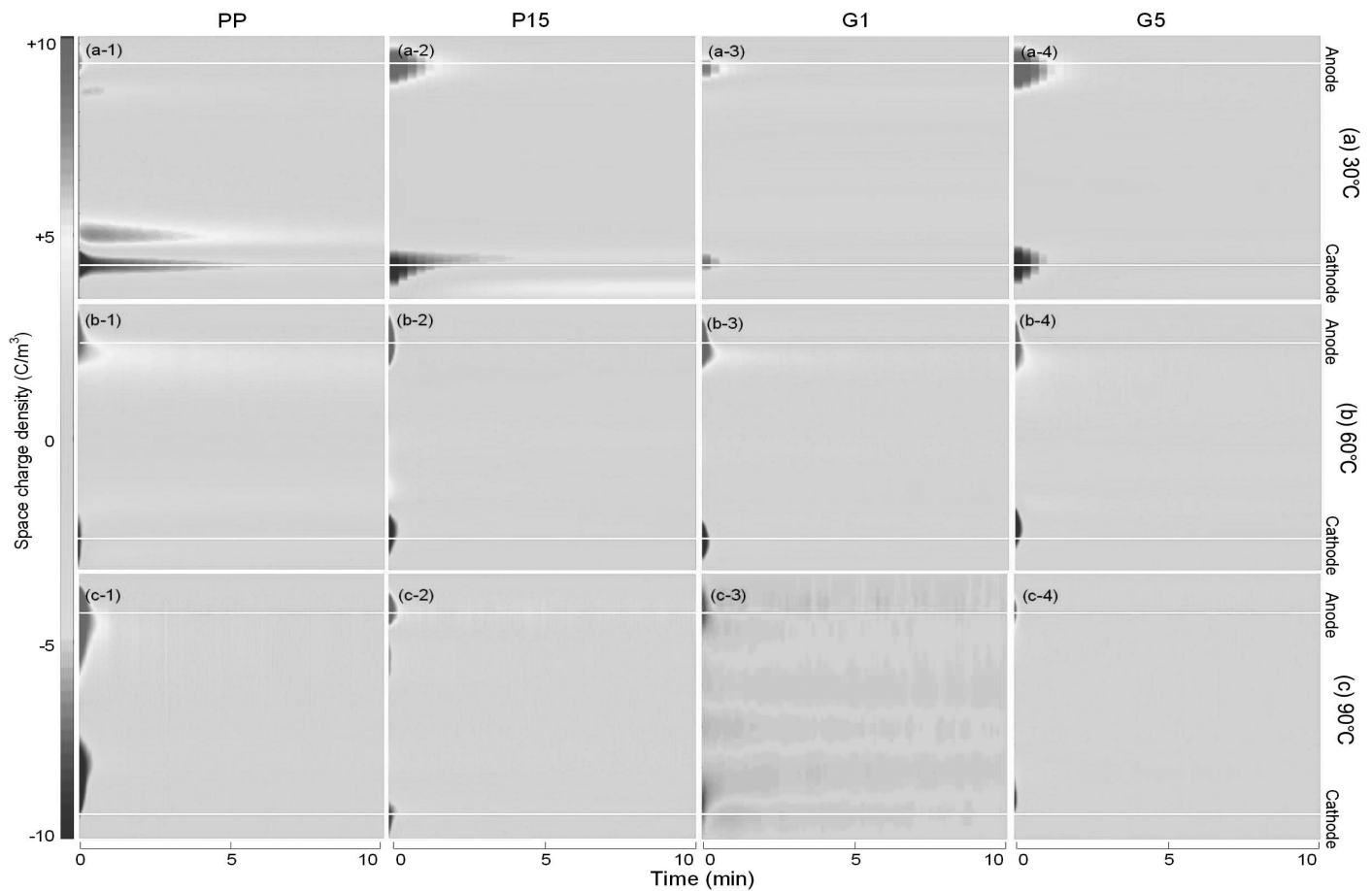


Figure 2. Space charge depolarization in PP and PP/ULDPE/graphene nanocomposites after the polarization of 60 min under 60 kV/mm at different temperature: (a) 30 °C, (b) 60 °C and (c) 90 °C.

In regard to P15 at 30 °C in Figure 2a-2, the space charge behavior during voltage-off process at different temperatures shows that the injected negative charge in the vicinity of the cathode decrease slowly. A small amount of hetero-charge can be found in the vicinity of both electrodes and the charges decay faster, which almost disappear after 1 min at 60 °C. In Figure 2c-2, negative charges appear near the anode and cathode electrodes and dissipate slowly with the voltage-off time at 90 °C.

For PP/ULDPE/graphene nanocomposites, G1 and G5, the space charge distribution in depolarization process shows different features. At 30 °C, there is a little amount of hetero-charge near the anode in G1 and a little amount of homo-charge in the vicinity of the cathode. Homo-charge can be found in the anode of G1 and negative charge can be observed in the bulk near the cathode of G5, where both the charges dissipate slowly at 60 °C. In terms of the 90 °C conditions of G1 and G5, there is almost no charge accumulated within the specimens from the very beginning of short circuit.

The mean volume density of space charge is introduced here to evaluate the relationship between space charge accumulation characteristics of different samples and test temperatures [22]. The mean volume density of space charge can be calculated according to the following equation:

$$q(t) = \frac{1}{d_2 - d_1} \int_{d_1}^{d_2} |\rho(x, t)| dx \quad (2)$$

where $q(t)$ is the mean volume density of space charge, d_1 and d_2 are the electrodes positions, t is the measurement time, $\rho(x, t)$ is the measured space charge density at position x and time t . The capacitive charge under field or induced charge at the electrodes in short circuit condition are not taken into account. The absolute value of charge density is used to provide cumulative data on the total amount of stored charge after 60 min polarization and 10 min de-polarization, respectively. The mean volume space charge density of different specimens at different temperatures after polarization for 60 min and de-polarization for 10 min are shown in Figure 3.

For the polarization test, as shown in Figure 3a, the mean charge density of PP, P15, G1 and G5 at 30 °C shows the lowest value in comparison to the conditions of 60 and 90 °C. With the increasing temperature, the density of all the samples were enhanced. PP shows the highest mean charge density at 90 °C and P15 show the almost same value at 60 and 90 °C. However, the G1 and G5 show the highest density value at 60 °C. In addition, the value of mean volume density at 30 and 60 °C shows an increasing trend with minimal fluctuations from PP to G5, but the value at 90 °C shows a downtrend. In the

case of depolarization, as shown in Figure 3b, the mean volume density shows various features. The highest value of PP to G5, are shown at 60, 90, 30 and 30 °C, respectively. For P15 and its blends with different contents of graphene, G1 and G5, the mean density shows three different variation trends.

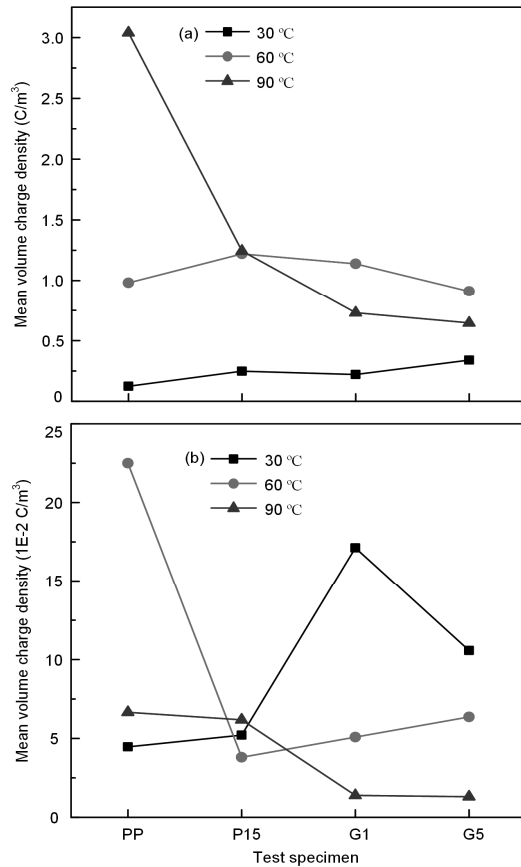


Figure 3. Relationship between mean volume charge density of different sample under 30, 60 and 90 °C: (a) polarization for 60 min and (b) depolarization for 10 min.

To further investigate the electric field distribution in different samples at 30, 60 and 90 °C, the electric field distribution in PP, P15, G1 and G5, after the polarization under 60 kV/mm for 60 min, is shown in Figure 4. The electric distortion factor can reflect the insulation ageing in somehow, which can be calculated from Equation (3):

$$D_E = \left| \frac{E_{MAX} - E_A}{E_A} \right| \times 100\% \quad (3)$$

where E_{MAX} is the highest electric field within the samples under 60 kV/mm for 60 min, E_A is the applied average electric field (60 kV/mm in this paper). The results are listed in Table 2. As shown in Figures 4a to 4c and Table 2, with the increasing temperature, the field distortion in the samples of PP, P15 and G5 has been strengthened. However, the distortion of G1 increases from 30 to 60 °C, then decreases from 60 to 90 °C, showing a higher value than that at 30 °C. P15 shows a relative higher distortion factor among the specimens at the same testing temperature.

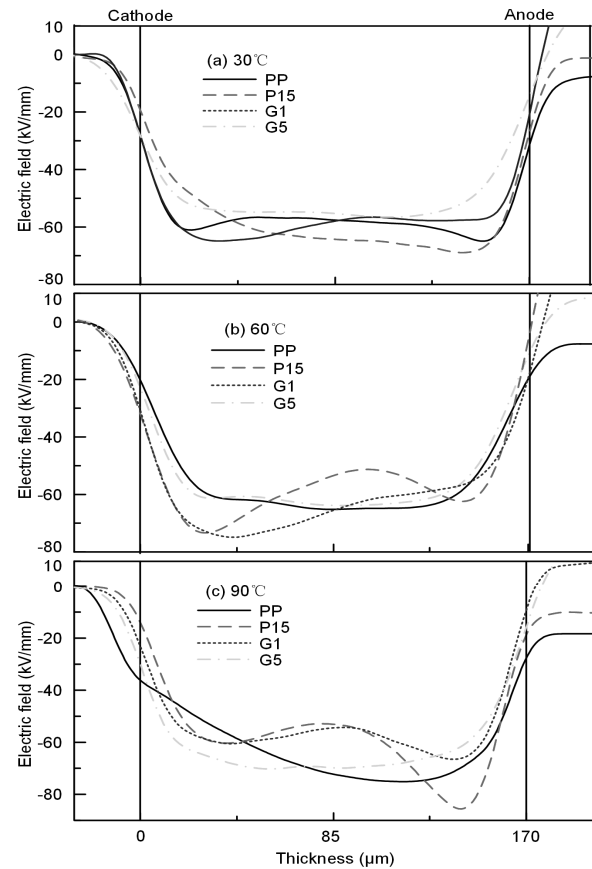


Figure 4. Electric field distribution in PP, P15, G1 and G5, after polarization under 60 kV/mm for 60 min at different temperature: (a) 30 °C, (b) 60 °C and (c) 90 °C.

Table 2. Maximum internal field distortion after polarization.

Temperature	PP	P15	G1	G5
30 °C	3.03	17.54	5.88	5.67
60 °C	8.50	22.13	24.65	6.30
90 °C	27.34	43.48	10.89	17.06

3.2 BREAKDOWN STRENGTH AND DC CONDUCTIVITY

The Weibull distribution of DC breakdown strength of PP/ULDPE/graphene nanocomposites is displayed in Figure 5. The 63.2 % breakdown strength of different specimens is listed in Table 3. As shown in Figure 5, different samples show various distributions of breakdown strength. The breakdown strength generally decreased with the increasing temperature. At the same time, G1 at 30 °C show the best performance among all of the samples and test temperature. On the other hand, G5 at 90 °C shows the worst performance with a bigger dispersion. In addition, most of the breakdown occurs between 250 and 350 kV/mm. As shown in Table 3, the breakdown performance at 30 °C reveals that the DC breakdown strength of PP has decreased slightly with addition of ULDPE, and the introduction of a low content of graphene (0.01 wt %) brings an increase of DC breakdown strength of the P15 blends, whereas the content of 0.05 wt.% led to an obvious decrease in G5. The similar conclusions can be made at 60 and 90 °C, except for P15 showing an improved performance at 60 °C.

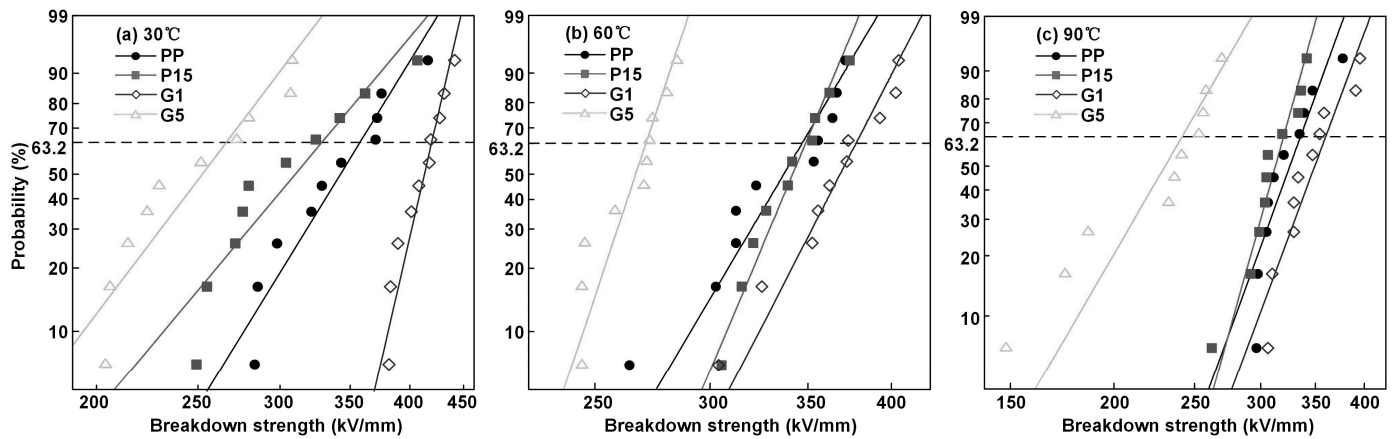


Figure 5. Weibull distribution of DC breakdown strength at different temperature: (a) 30 °C, (b) 60 °C, (c) 90 °C.

Table 3. 63.2 % breakdown strength at different temperatures.

Temperature	PP	P15	G1	G5
30 °C	358.9	329.2	420	267
60 °C	347.5	349.7	378.6	271.3
90 °C	335.6	320.2	359.5	241.6

For a further evaluation of temperature effects on charge conduction, the DC conductivity tests of different specimens are performed at 30, 60 and 90 °C, respectively, as depicted in Figure 6. It can be seen that, the conductivity of all the samples increases with the rising temperature, which has been largely improved at 90 °C. In terms of the same test temperature, the conductivity of the samples, from PP to G5, shows almost the same variation trend and G1 and G5 always show the lowest and highest value.

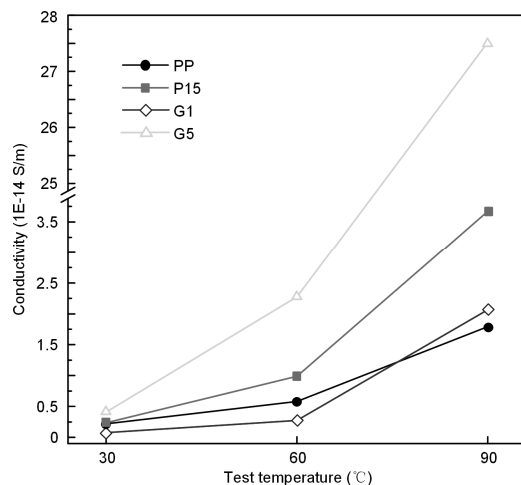


Figure 6. DC conductivity of different specimens at 30, 60 and 90 °C.

4 DISCUSSION

Space charge behavior in the polymer insulations are often accompanied by complex process of charge injection, transfer and extraction, especially when temperature is involved [7]. During this process, traps with a strong influence on the energy band of the molecule usually play an important role, which can be modified by additives, terminals and crystalline-

amorphous interfaces [8, 10, 17]. When the charge carriers transfer from one electrode to another, the carriers can be captured by traps, which can decrease homo-charge injection and charge free path length, restricting the charge mobility. At low temperatures, the injected or impurity ionized charges can be captured at shallow traps due to their relatively low energy and the space charge formation will be located near the electrodes [16]. At high temperatures, the accelerated charge mobility will make it more difficult for traps to trap the charges moving at high speed, because the trapped charges are thermally excited, easily reaching the required energy to de-trap [19,20]. Besides, the elimination and indistinctness of the interface between the phases, the density of local states, will decrease at high temperatures, and weaken the influence of traps on charges [15]. All these will result in the enhancement of charge injection and DC conductivity, with increasing temperature. All in all, the space charge distribution in polymer insulations is a dynamic balance of charge injection, transfer and extraction, at various temperatures.

The DC breakdown strength performance of semi-crystalline polymers can be affected by the testing conditions, the composition, morphology and secondary effect (space charge, etc.) [18]. When taking the temperature into account, it is worth noting that the temperature can not only affect the charge transfer and storage, but also change the physical states of polymer [17]. When the polymer exits in the state of rubber-like, the thermal or free volume breakdown should be taken seriously[16]. In the case of thermal breakdown, the amorphous region, especially the boundaries of spherulite are thought to be the weak point [18]. With the temperature keeping increase, the released segmental molecular motion and/or the increased rearrangement of free volumes can enhance the charge transport and charge energy in the polymer [16], which will increase the Joule heating and the probability of chain fracture [17], resulting in a lower breakdown strength. In addition, the accumulation of space charge in the bulk of insulation will result in electric field distortion, which is one of the main mechanism for insulation degradation, such as breakdown performance.

Bearing all this in mind, some characteristics of temperature influence on the space charge behaviors, DC conductivity and breakdown characteristics of PP/ULDPE/graphene nanocomposites can be interpreted as the following: as shown in Figure 1, a small amount of hetero-charge accumulated near the electrodes in PP at 30 °C, which is attributed to the high threshold value of charge injection for PP and some polarized impurities or ionized weakly bound under high DC electric field [5]. The increasing temperature has enhanced the charge injection, which is more dominant than the ionization. For pure PP, a higher charge injection at 60 °C than that at 30 °C and lower conductivity at 60 °C than that at 90 °C, result in a high mean charge density after de-polarization for 10 min.

As for P15 at 30 °C, the increased interfaces involved by blending with ULDPE bring more trap sites for the accumulation of space charge, and injected electrons neutralized the hetero-charges near the anode, as shown in Figures 1, 2 and 3. When it comes to P15 at 60 °C, the expanded ULDPE in PP make tighter PP/ULDPE phases and the reinforced spherulite boundaries result in an improved breakdown strength[24], as shown in Figure 5. As depicted in Figure 1, the injected charge can't get enough energy to over the barrier and extract to the electrodes, resulting in an enhancement of hetero-charges near both electrodes; thus it is deduced that the potential barrier for electron transport is higher at the anode than at the cathode.

Although a large amount of charges was injected into PP and P15 at 90 °C during the polarization, fewer charges remained in the sample, because of a high conductivity for charge dissipation at high temperature, as shown in Figures 2 and 6. However, the accumulated space charge after the polarization still create an electric field distortion in the bulk of the samples, as shown in Figure 4. P15 shows a worse distortion than PP at 90 °C, which is caused by the enhanced field between accumulated hetero-charge and anode, as shown in Figures 1 and 4. What's more, even if the dielectric breakdown field decreases with increasing temperature, no dramatic drop is observed at a critical temperature in this investigation. This is due to the absence of a transition phenomenon across the whole temperature range in this investigation [25] (T_g of PP and ULDPE is below 0 °C, T_m is above 100 °C) [6].

In terms of G1 from 30 to 90 °C, the space charge, breakdown strength and DC conductivity are enhanced by the introduction of graphene nanoplatelets in comparison of P15, which can be attributed to the inhibiting effects of the interaction zones at microcosmic interface between graphene and PP/ULDPE blends on the charge carrier transport [10-11]. As shown in Figures 1 and 3, less charge accumulated in G1 than P15 after polarization. The trapped space charge in G1 remains more than that in P15 after the de-polarization of 10 min at 30 and 60 °C. The mean charge density of G1 at 90 °C shows a smaller value than that in P15, which can be caused by a lower injection but a relative larger dissipation for G1. As shown in Figures 1 and 3, most of the charges are accumulated in the bulk of G1 and the charges of P15 exit in the vicinity of

the two electrodes, while they have almost the same mean charge density. The centralized charge distribution in G1 results in a larger field distortion than P15 at 60 °C, as depicted in Figure 4.

In comparison of P15, G1, it seems the space charge behavior of G5 has been improved with a small increase in graphene content, as depicted in Figures 1 to 4. However, the breakdown strength and DC conductivity indicate a worse dielectric property at different temperatures, shown in Figure 6. The cause of this phenomenon is mainly attributed to the high conductive paths for a local transport of charge carriers (not throughout) in the insulating matrix of PP/ULDPE blends [10-11]. It is considered that when the content of graphene is high enough, the polymer-filler interaction zones might overlap with each other, which provides more chance to form low-resistance paths for electrons and holes and accelerate a local transport of charge carriers (not throughout), thus resulting in the increase of DC conductivity and decrease of breakdown strength.

A low amount of graphene could involve deep trap sites, and a high content, such as G5, could introduce a greater chance of local low-resistance transport paths for carriers (not throughout) and result in higher conductivity. As shown in Figure 6, G5 shows the highest conductivity than PP and P15 as shown at 30, 60 and 90°C. The charges trapped by graphene in G5 are not easily de-trapped at low temperature (30°C), which results in a higher space charge accumulation and electric field distortion than PP and P15. At a higher temperature, 60 and 90°C, the injected charges with more energy are more difficult to be trapped and easier to de-trap. Thus, G5 brings about lower space charge accumulation and electric field distortion than PP and P15.

The mean charge density generally contains trapped and transporting charges under a relative stable state. At relative low temperatures, the charge injection and carriers' movement could be weakened. The higher the temperature, the more energy the charge carriers gain, and the more carriers are injected or dissociated, the higher of the carrier migration speed become, leading to an increased charge carrier density and mobility. As shown in Figure 3a and Figure 6, the mean volume charge density and the conductivity of different samples at 30°C shows a little change. When it comes to the 60°C, the conductivity increases a little higher despite almost the same level of mean volume charge density of the samples. However, when the mean volume charge density shows a decreasing trend, the conductivity of different samples at 90°C shows a larger improvement. As to the results above, it can be qualitatively inferred that the carrier density would play an important role in conductivity at 30°C, and at the elevated temperatures (60°C, and especially at 90°C) the mobility of carriers would play a dominant role.

5 CONCLUSIONS

The space charge behavior, breakdown strength and DC conductivity test of PP, PP/ULDPE and its graphene nanocomposites with different contents of 0.01 and 0.05 wt %

were performed at 30, 60 and 90 °C, respectively. From the experimental results and discussions presented above, the conclusions can be summarized as below:

(1) With the temperature increasing, the homo-charge injection from both electrodes in PP has been significantly enhanced. There're large amounts of electrons accumulated in PP/ULDPE blends near the vicinity of anode at 90 °C, which seems that negative charges are dominant carriers in PP/ULDPE blends at high temperature.

(2) Even if the breakdown strength decreases with increasing temperature, no dramatic drop has been observed, which could be due to the absence of a transition phenomenon of the blends at different temperatures in this investigation.

(3) The DC conductivity of all the samples increase with rising temperature, and the conductivity from 60 to 90 °C have a greater variation than that from 30 to 60 °C.

(4) The PP/ULDPE blends with a content of 0.01 wt.% graphene nanoplatelets show a good performance of space charge, DC conductivity and breakdown strength at the temperature range from 30 to 90 °C for potential HVDC cable insulation.

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